

IS 477 Interim Status Contribution Report – Colin Papp

For what we have accomplished so far, I was responsible for gathering both datasets, cleaning the season statistics dataset, and coming up with preliminary data visualizations. The steps I took and visualizations I created are outlined below.

Salary Data: from information science professional Edwin Jeon, who cleaned data from data scientist Erik Gregory Webb. Both individuals are full time data science professionals who have done this recreationally, and share their datasets for educational purposes, allowing free use, as stated in their Github pages. This dataset relevant columns we will use here are Player, Year, Salary, Position and Team. This is the dataset we will use to derive spending allocation for each team.

- Kaggle Link: <https://www.kaggle.com/datasets/ratin21/nba-player-stats-and-salaries-2000-2025?resource=download>
- Original Data: <https://github.com/erikgregorywebb/datasets/blob/master/nba-salaries.csv>

Cleaned Data: https://github.com/edwinjeon/NBA-Salary-Prediction/blob/main/data/NBA%20Player%20Stats%20and%20Salaries_2000-2025.csv

- Cleaning Methodology: <https://github.com/edwinjeon/NBA-Salary-Prediction/blob/main/NBA%20Salary%20Project%20Data%20Preparation.ipynb>

Statement of Free Use: <https://github.com/erikgregorywebb/data?tab=MIT-1-ov-fil>

Salary Data Preliminary Exploratory Data Analysis

- To explore trends in data before more computational work, we created visualizations to seek out trends in the data
- The first chart is a stacked bar chart of league-wide spending by position, by year
- The second chart is the first chart normalized since league spending on players has dramatically increased over time
- Both charts go back to 2000, but only 2005 and on will be used for further analysis
- We can see that a lower portion of player spending has gone towards centers and power forwards, with an increasing share going to guards. This can likely be attributed to the increasing importance of three point shooting in the league.

Season Data: from [craftedNBA.com](https://www.craftednba.com), a website for curated data for professional basketball fans. For our project, we used their NBA season data that spans back to 1955 and contains dozens of stats for each team, each season with stats including but not limited to Team Strength, Wins, Losses, Age (average age of players), SRS (strength of schedule impact), offensive/defensive ratings, true shooting, pace, and playoff wins and losses. Because we don't have a VIP membership to the website, we had to manually copy and paste data into the excel. We did not edit the excel, only used it as a medium to access the data after it was pasted in and converted to CSV for further analysis. This is the dataset we will use to compare against spending allocation.

- [Season Data.xlsx](#)
- [Historical NBA Team Stats | CraftedNBA | NBA Stats & Analytics](#)
- The dataset is comprised of factual, non-copyrightable NBA statistical information (wins, losses, efficiency metrics) and is being used solely for non-commercial analytical and educational purposes

Season Data Processing:

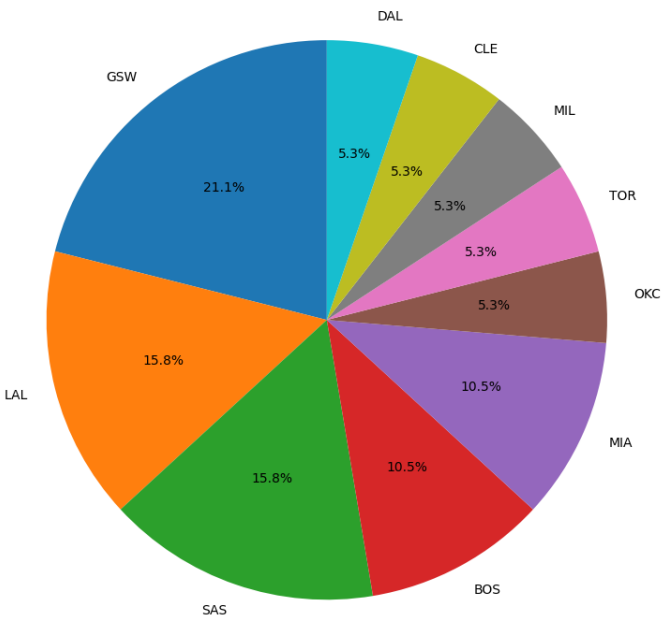
- The data was filtered to exclude most of the playoff advanced stats, only including playoff wins and losses, it was also filtered to only include 2005 and beyond.
- Blank playoff wins and losses column were filled with zeros, as these columns were intended to be left blank since the teams missed the playoffs.
- A boolean 0/1 variable was created for both playoff appearance and championship win, if a season had playoff wins/losses listed it was marked as a playoff appearance. From the initial data, it had an emoji by the team's name if they won that season which was read as question marks, so if a team had question marks in their name they were marked as winning the championship and the question marks were removed.
- Since teams enter and exit the NBA, the data was filtered for teams that have at least 17 seasons represented in the dataset
- The data was checked for na values, and none remained
- The salary dataset used abbreviated team names while the season data used full names (e.g. CLE instead vs. Cleveland Cavaliers). To fix this, a name map was created matching the full names to the abbreviated versions, and team names in the season dataset were mapped to match the names in the salaries dataset

Season Dataset Preliminary Exploratory Data Analysis

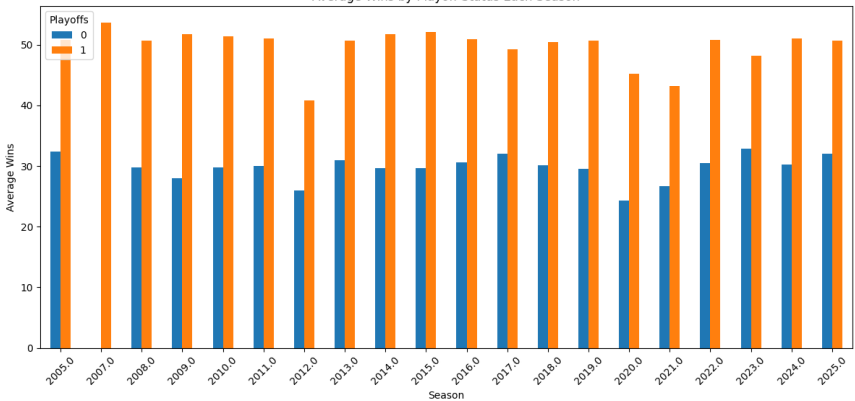
- For the purpose of exploratory data analysis, we created the following graphs using Matplotlib to see the distribution of championships, average wins for playoff vs. non playoff teams by year, and average win percentages for teams across the dataset. This was done using `value_counts()`, `.groupby()`, and creating a win percentage column

Visualizations Created:

Distribution of Championship Winning Teams



Average Wins by Playoff Status Each Season



Average Win Percentage by Team Across All Seasons

